

DmV Calculator PRO

Complete User Manual

Advanced Scientific Calculator for Android

Four Modes · Unit Converter · Persistent History · Full Customization

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DmV Calculator PRO is a professional-grade scientific calculator for Android, engineered for engineers, scientists, students, and power users who demand accuracy, speed, and versatility in a single compact application. With four distinct operating modes, a comprehensive built-in unit converter, a fully persistent session history, and seamless multi-window support, DmV Calculator PRO redefines what a mobile calculator can do.

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1. USER INTERFACE OVERVIEW

The DmV Calculator PRO screen is organized into logical zones: the **Action Log** (history panel) at the top, the **Notification Bar** below the log, the **Undo/Redo** controls, the **Input Field**, the **Result Preview**, and the **Keypad**. All zones scale dynamically to any window size, including split-screen and floating pop-up views.

Action Log — Scrollable history panel showing all past expressions and results with session timestamps. Persists across app restarts.

Notification Bar — A dedicated message strip located between the Action Log and the Undo/Redo row (portrait modes) or to the right of the result field above the converter grid (Full Function mode). Displays input-error alerts, hints, tips, and button/category labels on long-press. See Section 4 for full details.

Undo ↶ / Redo ↷ — Two circular-arrow icons. Step backward or forward through recent edits to the current expression.

Input Field — The active expression editor. Supports full text editing: tap to position cursor, long-press to select, copy, cut, or paste.

Result Preview — The large green number below the input field shows the live calculated result. Tap it to instantly copy the value to the clipboard.

Top Bar — i button — Displays brief help text and a summary of available shortcuts.

Top Bar — ≡ (Menu) — Opens the settings menu to switch modes, toggle themes, or save the current layout.

Top Bar — clear — Clears the entire Action Log (visible when log is non-empty).

► **Pro Tip:** Tap the large green result number to copy it to the clipboard. A 'Copied.' toast confirms the action.

2. CORE KEYPAD — BUTTONS & SHORTCUTS

The following keys are present in all operating modes:

Key / Button	Function
=	Calculate the current expression and append it to the Action Log.
C	Cancel / Clear — erases the current input without affecting the Action Log.
<x	Backspace — deletes the last character in the input field.

R	Last Result — inserts the numeric result of the most recent calculation into the input field at the cursor position. Ideal for chaining calculations.
AL	Toggle Action Log — shows or hides the scrollable history panel.
.	Decimal point (tap). Long-press to enter E-format scientific notation (e.g. $0.45225E5 = 45225$).
()	Parentheses for grouping sub-expressions.
%	Modulo — remainder after division.
x[^]	Power / exponentiation (e.g. $2x^{10} = 1024$).
√	Square root.
∛	Cube root (Engineer & Full Function modes).
 x 	Absolute value (Engineer & Full Function modes).
<>	Swap converter variables — inverts the unit conversion direction (Converter & Full Function modes).

► **Pro Tip:** Long-press the decimal point '.' key to switch a number segment into E-format scientific notation. Useful for very large or very small values.

3. OPERATING MODES & NAVIGATION

DmV Calculator PRO provides four distinct layouts. Switch between them at any time via the **≡ top menu** or by swiping left/right across the keypad area (see Section 3.5).

3.1 Simple Mode

A clean, distraction-free layout with standard arithmetic operators: addition, subtraction, multiplication, division, percentage (%), powers (x[^]), and square root (√). Designed for fast everyday calculations.

3.2 Engineer Mode

Expands the keypad with a full suite of advanced mathematical functions:

- Trigonometry: **sin**, **cos**, **tan** (all angles in **degrees**)
- Inverse trig: **asin**, **acos**, **atan**
- Hyperbolic: **sinh**, **cosh**, **tanh**
- Logarithmic: **ln** (natural log), **lg** (log base 10)

- Constants: π (pi), e (Euler's number)
- Cube root: $\sqrt[3]{}$
- Absolute value: $|x|$

Important: All trigonometric functions operate strictly in **degrees**.

3.3 Converter Mode

Provides a comprehensive grid of unit conversion buttons organized by category. The standard arithmetic keypad is retained alongside the converter grid.

1. Enter a numeric value in the input field.
2. Tap the **source unit** button (e.g. **lb**).
3. Tap the **target unit** button (e.g. **kg**). The expression (e.g. **4.48 lb→kg**) appears and the result is shown instantly.
4. Use **<>** to swap source and target units.

3.4 Full Function Mode

Combines the Engineer Mode keypad and the complete Unit Converter on a single screen, giving simultaneous access to all scientific functions and all conversion categories without switching modes. In this layout the Notification Bar is positioned to the right of the result field, directly above the converter grid.

3.5 Switching Modes: Menu & Swipe Gestures

DmV Calculator PRO offers two ways to switch between modes:

- **≡ Settings Menu** — tap the menu icon in the top-right corner of the title bar. A dropdown appears with all four mode options (SIMPLE, ENGINEER, CONVERTER, FULL FUNCTION) plus theme and Save Mode controls. Tap any mode name to switch immediately.
- **Swipe gesture** — swipe **left or right** across the keypad area to cycle through modes in sequence. This is the fastest way to switch without lifting your thumb to the top bar.

► **Pro Tip:** Swipe right to advance to the next mode (Simple → Engineer → Converter → Full Function) and swipe left to go back. Once you've found your preferred layout, use **SAVE MODE** so the app always opens in it.

4. NOTIFICATION BAR

The Notification Bar is a dedicated message strip built into every operating mode. It provides real-time feedback without interrupting your workflow.

Position by layout

- **Simple / Engineer / Converter modes (portrait)** — displayed between the Action Log and the Undo/Redo row, directly above the input field.

• **Full Function mode** — displayed to the right of the large result field, above the unit converter grid.

What the Notification Bar shows

Input error alerts — Whenever an expression contains a syntax error, mismatched parentheses, or an illegal operation (e.g. division by zero), the bar immediately displays a descriptive error message so you can correct the input before pressing =.

Operation hints — Context-sensitive tips appear as you build expressions, guiding you toward correct syntax for complex functions.

Button labels on long-press — Long-pressing any key on the keypad displays the full name or description of that button in the Notification Bar (e.g. long-pressing **tanh** shows ‘Hyperbolic tangent’). Useful for discovering the exact function behind an unfamiliar button.

Converter category labels on long-press — Long-pressing any category header or unit label in the converter grid displays its full name and a brief description in the Notification Bar (e.g. long-pressing **ct** shows ‘Carat — mass’).

Swap confirmation — After pressing **<>** to swap conversion direction, the bar confirms the new conversion pair (e.g. ‘lb ↔ ct swapped’).

General tips & advice — Occasional usage tips appear in the bar to help you discover less obvious features of the application.

i Note: The Notification Bar is read-only and updates automatically. No interaction is needed — simply glance at it whenever you need guidance.

5. UNIT CONVERTER — COMPLETE REFERENCE

The built-in converter covers 8 measurement categories with over 40 individual units. Conversions are integrated into the expression engine and appear in the Action Log like any other calculation, fully reusable with the **R** key.

Category	Available Units
Mass	ton, tn, lb, oz, mt, kg, g, ct, mg
Length	nm, mi, yd, ft, in, km, m, cm, mm
Area	mi², acre, yd², ft², in², km², ha, m², cm²
Volume	yd³, ft³, in³, gal, usgal, pt, uspt, li, m³, cm³, ml
Pressure	bar, mmhg, atm, psi, kpa

Speed	mph, fps, kph, mps, kn
Energy	btu, ev, kj, kwh
Temperature	°F, °C, K

► **Pro Tip:** After a conversion, press **R** to insert the converted result into a new expression. Chain multiple conversions or feed the result directly into a scientific function — for example, calculating the sine of a converted angle.

6. ACTION LOG — PERSISTENT SESSION HISTORY

The Action Log is an interactive calculation tape that persists between sessions. Your entire history — expressions, results, timestamps — is saved automatically and restored on every launch.

Persistent storage — All history is saved automatically. Closing and reopening the app restores the full log exactly as you left it.

Session timestamps — Each new session is marked with a **-New date-** entry (e.g. -New date- 18-Jun-2026 20:30:12), making it easy to locate calculations from a specific date or work session.

Double-tap to select — Double-tap any expression in the log to highlight it.

Tap to insert — Tap a highlighted expression to insert it directly into the current input field. Recall any past formula instantly without retyping.

Long-press to select text — Long-press anywhere in the log to enter text-selection mode. Select, copy, or share any portion of your history.

Toggle visibility — Press **AL** to hide or show the log, freeing up screen space for the keypad.

Clear log — Tap **clear** in the top bar to erase the entire Action Log permanently.

► **Pro Tip:** Use session timestamps to organize your work across multiple days. Each app launch adds a new timestamp marker, making it easy to separate, review, or share results from different sessions.

7. INPUT FIELD & EXPRESSION EDITING

The input field is a full text editor. It supports all standard Android text-editing gestures.

- Tap anywhere in the field to position the text cursor.
- Use **<x** to delete characters one at a time.

- Long-press to open Android text-selection handles for selecting, cutting, copying, and pasting.
- Use **Undo** ↶ / **Redo** ↷ to step through the edit history.
- The **R** key inserts the last computed result at the current cursor position.
- Tapping a highlighted Action Log entry appends its expression at the cursor position.
- Expressions support full nesting: parentheses, nested functions (e.g. **$\text{abs}(\sqrt[3]{\text{atan}(x)})$**), and chained operations.

8. SETTINGS & CUSTOMIZATION

All settings are accessed through the ≡ **top-right menu**.

SIMPLE / ENGINEER / CONVERTER / FULL FUNCTION — Tap any mode name to switch layouts immediately. Alternatively, swipe left or right on the keypad.

DARK / LIGHT — Toggle between dark and light themes. Your choice is remembered and applied automatically on every subsequent launch.

SAVE MODE — Saves the currently active operating mode as the default startup configuration. On next launch the app opens directly in this saved mode. Theme preference is also saved automatically.

► **Pro Tip:** Set FULL FUNCTION as your SAVE MODE default for the fastest workflow: every scientific function and every unit conversion category is available the instant you open the app.

9. MULTI-WINDOW, FOLDABLES & TABLET SUPPORT

DmV Calculator PRO fully supports all Android multi-window display modes and adapts seamlessly to any screen form factor:

Full-screen mode — Standard operation with maximum display area.

Split-screen mode — Run side-by-side with another app. The UI reflows automatically to fit the reduced width, keeping all buttons and the log accessible.

Pop-up (floating window) mode — Launch as a small floating overlay on top of any other app. Ideal for quick lookups while working in a document or browser. Resize and reposition the window freely.

Foldable phones (Galaxy Z Fold and similar) — DmV Calculator PRO works correctly in both folded (cover screen) and unfolded (large inner display) states. The interface scales dynamically when the device is opened or closed, making full use of the expanded screen real estate in unfolded mode.

Tablets — On tablets the larger display area is used efficiently: the keypad, Action Log, and converter grid spread out comfortably, reducing the need for scrolling and making all

controls easier to tap. Full Function mode is especially well-suited to tablet use.

In all window modes and on all device types the Action Log remains fully functional and persistent history continues recording every calculation. No functionality is lost regardless of window size or screen form factor.

10. QUICK-REFERENCE CARD

Action	How to do it
Calculate expression	Press =
Clear current input	Press C
Delete last character	Press <x
Insert last result	Press R
Enter scientific notation	Long-press the . (dot) key
Toggle history panel	Press AL
Copy result to clipboard	Tap the large green result number
Recall past expression	Double-tap in log, then tap to insert
Select text in log	Long-press in the Action Log
Convert units	Enter value → tap source unit → tap target unit
Swap conversion direction	Press <>
Switch mode (menu)	≡ menu → choose mode
Switch mode (gesture)	Swipe left or right on the keypad
Toggle dark/light theme	≡ menu → DARK / LIGHT
Save startup mode	≡ menu → SAVE MODE
Undo / Redo input edit	Tap ↶ or ↷ above the input field
See button name	Long-press any key → Notification Bar
See converter unit name	Long-press unit label → Notification Bar